Machine Learning Model Results Report

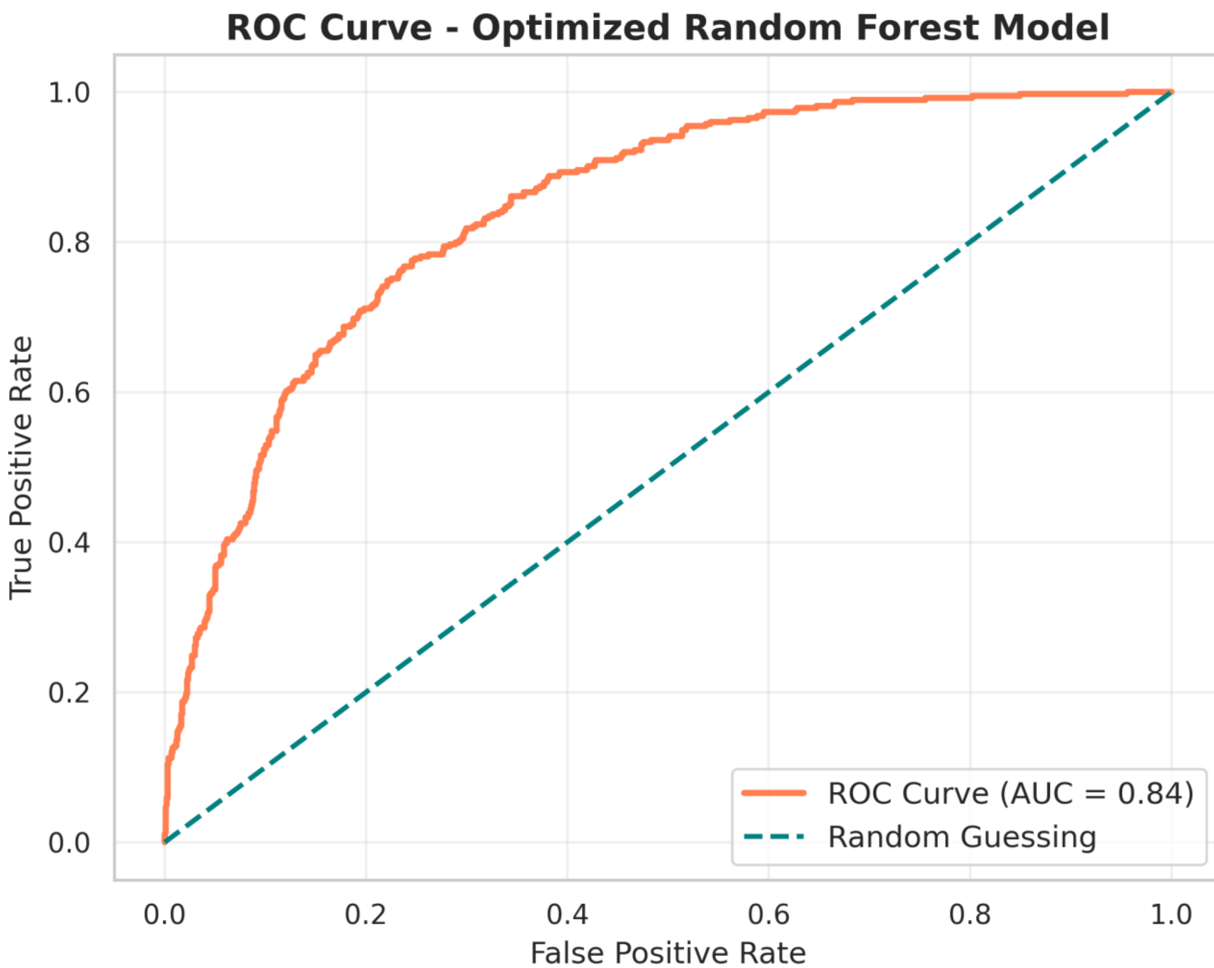
1. Overview

The primary objective of this phase was to develop a robust machine learning pipeline to predict customer churn. Two models were trained, evaluated, and compared using multiple classification metrics to ensure accuracy and reliability.

2. Model 1: Random Forest Classifier

This model was tuned using a maximum depth limit (`max_depth=8`) to prevent overfitting while effectively capturing complex patterns in customer behavior.

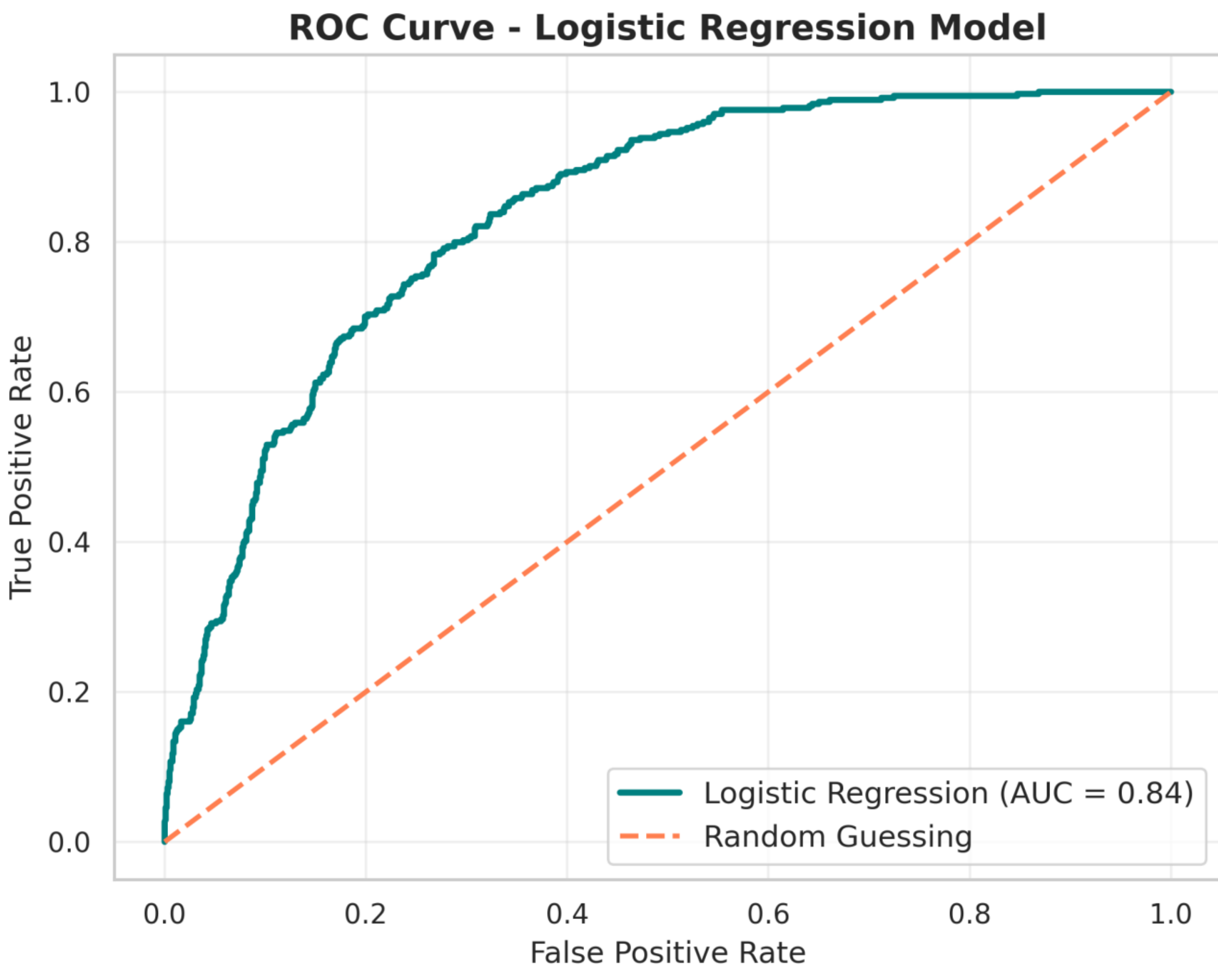
- **Overall Accuracy: 79.91%**
- **Precision (Class 1 - Churn): 0.66**
  - *Explanation:* Out of all customers the model predicted would churn, 66% actually did.
- **Recall (Class 1 - Churn): 0.51**
  - *Explanation:* Out of all actual churning customers, the model successfully identified 51%.
- **F1-Score (Class 1 - Churn): 0.57**
  - *Explanation:* The balanced harmonic mean of Precision and Recall.
- **ROC-AUC Score: 0.84**



3. Model 2: Logistic Regression

The dataset features were scaled using `StandardScaler` to ensure optimal performance for this linear classification algorithm.

- **Overall Accuracy: 79.56%**
- **Precision (Class 1 - Churn): 0.63**
- **Recall (Class 1 - Churn): 0.55**
- **F1-Score (Class 1 - Churn): 0.59**
- **ROC-AUC Score: 0.8353**



4. Conclusion

Both models demonstrated robust predictive capabilities, successfully identifying patterns in customer attrition with accuracies nearing 80%.

While Logistic Regression provided a slightly better Recall (0.55)—meaning it caught a few more actual churners—it did so at the cost of more false alarms (lower precision). On the other hand, the **Optimized Random Forest** outperformed Logistic Regression in overall Accuracy (79.91%), Precision (0.66), and ROC-AUC (0.84). This means the Random Forest model is much more reliable at correctly identifying true churn risks without unnecessarily flagging loyal customers, which helps in optimizing the budget for retention campaigns.

Recommendation: The Optimized Random Forest model is recommended as the primary prediction engine for deployment. Its superior precision and ability to handle complex, non-linear customer behaviors make it a highly cost-effective and accurate tool for driving proactive retention strategies.